

EOSnet: Embedded Overlap Structures for Graph Neural Networks in Predicting Material Properties

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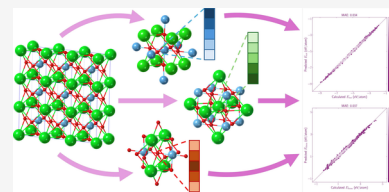


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ABSTRACT: Graph Neural Networks (GNNs) have emerged as powerful tools for predicting material properties, yet they often struggle to capture many-body interactions and require extensive manual feature engineering. Here, we present EOSnet (Embedded Overlap Structures for Graph Neural Networks), a novel approach that addresses these limitations by incorporating Gaussian Overlap Matrix (GOM) fingerprints as node features within the GNN architecture. Unlike models that rely on explicit angular terms or human-engineered features, EOSnet efficiently encodes many-body interactions through orbital overlap matrices, providing a rotationally invariant and transferable representation of atomic environments. The model demonstrates superior performance across various prediction tasks of materials' properties, achieving particularly notable results in properties sensitive to many-body interactions. For band gap prediction, EOSnet achieves a mean absolute error of 0.163 eV, surpassing previous state-of-the-art models. The model also excels in predicting mechanical properties and classifying materials, with 97.7% accuracy in metal/nonmetal classification. These results demonstrate that embedding GOM fingerprints into node features enhances the ability of GNNs to capture complex atomic interactions, making EOSnet a powerful tool for materials' discovery and property prediction.



Machine learning (ML) has become an indispensable tool in the fields of materials science and condensed matter physics, enabling the efficient prediction and discovery of material properties that would otherwise be time-consuming using first-principles calculations. As the demand for high-throughput materials screening has grown, ML models, particularly deep learning architectures,^{1,2} have evolved to handle increasingly complex data sets, capturing intricate relationships between atomic structures and their properties. Within this context, Graph Neural Networks (GNNs)^{3,4} have emerged as particularly well-suited for modeling atomic and molecular systems due to their ability to process graph-based data, where atoms and their interactions are represented as nodes and edges, respectively.

In the early applications of ML to materials science, models such as high-dimensional neural networks (HDNNs)⁵ were employed to learn the complex potential energy surface (PES) across large atomic systems. These models utilized various descriptors like Atom-Centered Symmetry Functions (ACSFs),^{6,7} SO(3) power spectrum or Smooth Overlap of Atomic Positions (SOAP),^{8,9} SO(4) bispectrum or Spectral Neighbor Analysis Potential (SNAP),¹⁰ and Atomic Cluster Expansion (ACE).^{11–14} Each descriptor had its own trade-offs. For example, ACSFs required parameter tuning,¹⁵ while SOAP and SNAP were computationally intensive due to spherical harmonics and higher-order angular terms.^{16,17} ACE provided a complete, scalable framework for atomic representation but introduced challenges with higher-order term computations.¹⁸

HDNNs demonstrated near-DFT accuracy in predicting energy-related properties for elemental crystals with simple

lattice types. However, these models require individual training for specific chemical systems and struggle with binary, ternary, or quaternary systems featuring complex lattices, limiting their transferability. Additionally, all four descriptors focused on local atomic environments within a cutoff radius, lacking a global encoding of the entire systemic geometry and topology. Efforts to incorporate long-range interactions, such as electrostatics^{2,19} and nonlocal charge-transfer,^{20–23} has greatly improved HDNN accuracy but still primarily targeted DFT energies and partial charges. Despite these advancements, accurate prediction of electronic properties, such as Fermi energy and band gap, remains an area requiring further exploration.

GNNs, on the other hand, eliminate the need for extensive descriptor engineering by directly learning from structural data. This capability enables GNNs to model both the local atomic environment and the global structure of materials in a more natural and scalable manner, making them particularly useful for applications in materials discovery, such as predicting electronic properties, phase transitions, and mechanical properties.^{24–30} One of the most influential models in this area is the Crystal Graph Convolutional Neural Network

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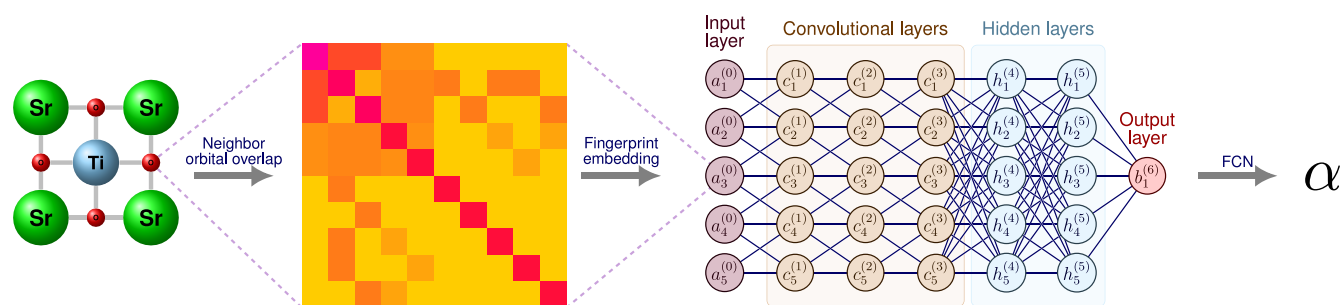


Figure 1. Schematic architecture of EOSNet, demonstrating how GOM Fingerprints are embedded into the Graph Convolutional Neural Network (GCNN) framework. Here we use the SrTiO₃ crystal structure (top-down view) as an example. Each atom in the crystal are used to build the atom-centered Gaussian Overlap Matrix (GOM) that captures neighbor orbital overlaps. A normalized GOM for an O atom is shown as an example. The diagonal elements represent self-overlap of atoms, with the (1,1) element corresponding to the central atom self-overlap, which is maximal due to the smooth cutoff function f_c used in GOM construction. The first row/column shows overlaps between the central atom and its neighbors, while other off-diagonal elements show overlaps between neighbors. The GOM eigenvalues are input as original node features (purple) for the GCNN. Then the CGNN learn the hidden node features by feature aggregation and message-passing (see details in Figure 2) in the graph convolution (orange). Then hidden layers (cyan) are used to extract hidden features of each node then use average pooling to pool the node features to crystal feature (pink). Finally we use on last fully connected layer (FCN) to produce the predicted material property, denoted by α .

(CGCNN),³¹ introduced by Xie and Grossman in 2018. CGCNN represents materials as graphs to capture the structural and bonding information on crystalline solids. This representation, combined with convolutional layers that update atomic and bond features through a message passing mechanism,³² makes CGCNN highly effective in predicting material properties from crystal structures. The success of CGCNN sparked widespread interest in the use of GNNs for material property prediction.

Despite the success, distance-based message-passing GNN architectures like CGCNN face two major limitations. One limitation is that they only encode pairwise distance information for edge features, neglecting many-body interactions that are essential for accurately modeling atomic environments.³³ Many-body interactions involve complex dependencies among multiple atoms, where the properties of an atom cannot be fully described by its interactions with just one or two neighbors. Another challenge is that current GNN-based machine learning models often require extensive human intervention and numerous *in silico* experiments to select appropriate chemical information (such as atomic number, valence electrons, electronegativity, covalent radius) for node feature encoding. This manual feature selection limits the model performance and transferability, particularly when working with smaller data sets.^{34,35}

To address the first limitation, researchers have enhanced GNN architectures by incorporating extra geometric information and improving how these models process atomic interactions. One key improvement involves integrating more complex geometric features beyond simple bond lengths, such as angular and directional information. Models like DimeNet,³⁶ GemNet,³⁷ ALIGNN,³⁸ and M3GNet³⁹ leverage these geometric details to capture intricate spatial relationships between atoms, which is particularly important in anisotropic materials. Moreover, models like E3NN²⁷ and NequIP⁴⁰ incorporate vector information to ensure the representations remain invariant under rotations and translations, thereby enhancing the ability to model complex atomic systems. In addition, attention mechanisms and global aggregation techniques have revolutionized how GNNs handle information flow. Attention layers, used in models like GATGNN⁴¹ and Equiformer,⁴² dynamically adjust the importance of different atomic interactions, allowing the network to emphasize the most

chemically relevant bonds. These attention mechanisms enhance local expressiveness and improve the aggregation of global structural information. By combining these features with advanced readout functions—such as global attention and hierarchical pooling—models like MEGNet⁴³ and GraphTrans⁴⁴ ensure that critical information is preserved across the entire structure. Together, these innovations allow GNNs to efficiently prioritize and synthesize both local and global atomic interactions, improving the accuracy of material property predictions. Beyond these general developments, there have also been efforts to tailor GNNs specifically for crystalline materials. For example, GeoCGNN⁴⁵ and Mat-former²⁸ explicitly encode crystal periodicity, while the Reciprocal Space Neural Network (RSNN)⁴⁶ leverages reciprocal space information to capture long-range interactions in periodic systems. These developments underscore the growing complexity and sophistication of GNN architectures in materials science, enabling more accurate and generalizable property predictions. Recently, MACE^{47,48} has been proposed as another advanced framework that systematically incorporates many-body interactions into message-passing neural networks by employing high-order tensor products of spherical harmonics. MACE has demonstrated state-of-the-art performance for atomic-scale simulations, primarily excelling in fitting potential energy surfaces for molecules and materials.

In this work, we introduce EOSNet (Embedded Overlap Structures for Graph Neural Networks) to overcome the limitations in GNN models for materials science. EOSNet adopts a new approach to incorporate many-body interactions through Gaussian Overlap Matrix (GOM) fingerprints^{49,50} as atomic features within the GNN architecture. Unlike previous GNN models that embed angular information into edge features explicitly, or rely on spherical harmonics expansions and careful truncation to achieve rotational invariance and many-body completeness (as in MACE,^{47,48} which is mainly for fitting potential energy surfaces), EOSNet focuses on predicting material properties, such as electronic band gaps. By representing the overlap of atomic orbitals between neighboring atoms through GOMs, EOSNet provides a compact and efficient description of the many-body interactions that govern these properties. Moreover, the fingerprint features in EOSNet are generic and transferable, ensuring that two atoms share the same node feature only if they have identical neighboring

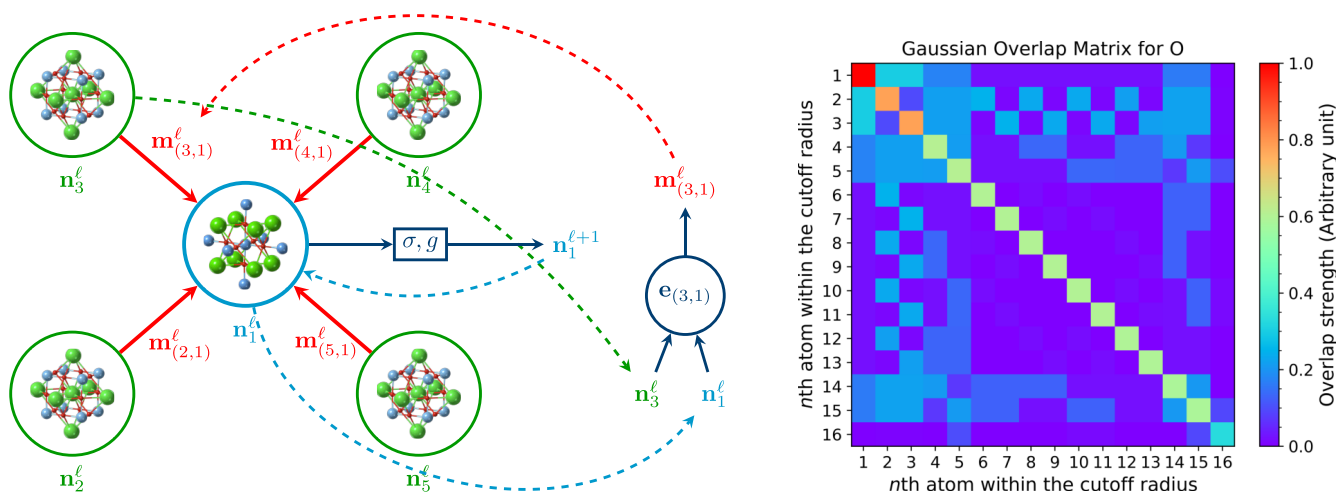


Figure 2. Illustration of many-body interaction and GOM. Left: Demonstration of the feature aggregation and message-passing scheme for an SrTiO₃ crystal using GOM fingerprints. Green, blue, and red spheres represent Sr, Ti, and O atoms, respectively. For simplicity, only four Sr atoms are considered as nearest neighbors to the Ti atom. The $\mathbf{m}_{(ij)}^l$ is the message from atom j to atom i denoted in the Equation 2 in the main article. $\mathbf{n}_i^l$ denotes the node feature of atom i at layer l , $\mathbf{e}_{(ij)}$ represent the edge feature between atom i and j from Equation 1 to (2). Right: Normalized Gaussian Overlap Matrix for an O atom, with atoms within the cutoff radius sorted by their distance from the central atom. The diagonal elements indicates the self-overlapping, and the first entry of the diagonal elements correspond to the central atom. For demonstration purposes, we only show the first 16 GOM overlap elements with s orbitals.

environments, even within the same atomic species. Another advantage of the GOM-based fingerprints is their inherent rotational and translational invariance, ensuring a consistent description of atomic environments regardless of structural orientation or position. By embedding these fingerprints as node features, EOSnet achieves superior predictive performance across a range of material property tasks, particularly those where orbital overlaps are critical, while maintaining computational efficiency.

In EOSnet, the crystal structure is represented as a graph, where each atom is a node, and the edges represent bonds between neighboring atoms. The model architecture, depicted in Figure 1, integrates GOM-based fingerprints within a graph convolutional neural network (GCNN) framework. These GOM fingerprints are derived from the eigenvalues of Gaussian overlap matrices, designed to capture many-body atomic interactions within a cutoff sphere centered around each atom in the crystal structure.

To construct the GOM, Gaussian-type orbitals (GTOs) are centered on each atom within this sphere, and the overlap integrals are calculated between every pair of atoms within the cutoff radius. The width of each Gaussian function is determined by the covalent radius of the atom on which it is centered, incorporating key features of the covalent radius into the atomic representation. Additionally, a cutoff function ensures that the overlap integrals smoothly decay to zero at the boundary of the sphere, preventing discontinuities when atoms enter or leave the region. The definitions of the GOM matrix elements and the smooth cutoff function are given by

$$[\mathbf{O}_m]_{i+l,j+l'} = f_c(r_{im}) \langle \phi_i^l | \phi_j^{l'} \rangle f_c(r_{jm}) \quad (1a)$$

$$f_c(r) = \left(1 - \frac{r^2}{r_{\text{cut}}^2}\right)^n \quad (1b)$$

where $\mathbf{O}_m$ represents the GOM for atom m in the unit cell, and f_c is the smooth cutoff function, defined by a user-specified cutoff radius r_{cut} and a nonlinearity parameter n . Indices i and j

label the atoms inside the cutoff sphere, and $r_{im} = |\mathbf{r}_i - \mathbf{r}_m|$ and $r_{jm} = |\mathbf{r}_j - \mathbf{r}_m|$ denote the distances between atom m and atoms i and j , respectively. The orbital indices l and l' correspond to specific angular momentum states. The Gaussian-type orbitals (GTOs) are given by

$$\phi_i^l(\mathbf{r}) = N_i (x - x_i)^{l_x} (y - y_i)^{l_y} (z - z_i)^{l_z} e^{-\alpha_i |\mathbf{r} - \mathbf{r}_i|^2}$$

where N_i is the normalization factor and $\alpha_i = 0.5(r_i^{\text{cov}})^{-2}$, with r_i^{cov} being the covalent radius of atom i .

The fingerprint for each atom is then computed by extracting the eigenvalues of its corresponding GOM. These eigenvalues encode essential information about atomic environments and serve as rotation and translation invariant features for the graph neural network. This design captures the strength of interactions between an atom and its neighbors, as well as the interactions among the neighboring atoms themselves, offering a comprehensive and efficient representation of many-body atomic interactions. Since the Gaussian parameters used in the GTOs are fixed, the descriptor construction process is simplified, avoiding extra tuning for specific systems. For moderate-sized systems, GOM-based fingerprints offers a balance between expressiveness and computational efficiency. Unlike methods that rely on costly angular expansions, such as SOAP and bispectrum descriptors, GOMs produce a matrix (or eigenvalue) representation that can be directly fed into machine learning models without extensive preprocessing. This straightforward mathematical structure allows neural networks to capture both local and global structural properties effectively. Figure 2 illustrates the message-passing mechanism of EOSnet and the many-body interaction nature of the GOM fingerprints in a cubic SrTiO₃ structure, illustrating how the overlap strength between atomic orbitals is encoded in the GOM.

The EOSnet architecture follows a standard message-passing framework, where the node features (GOM fingerprints) are updated based on the interactions between neighboring atoms. The workflow of EOSnet is illustrated in Figure 3. In each

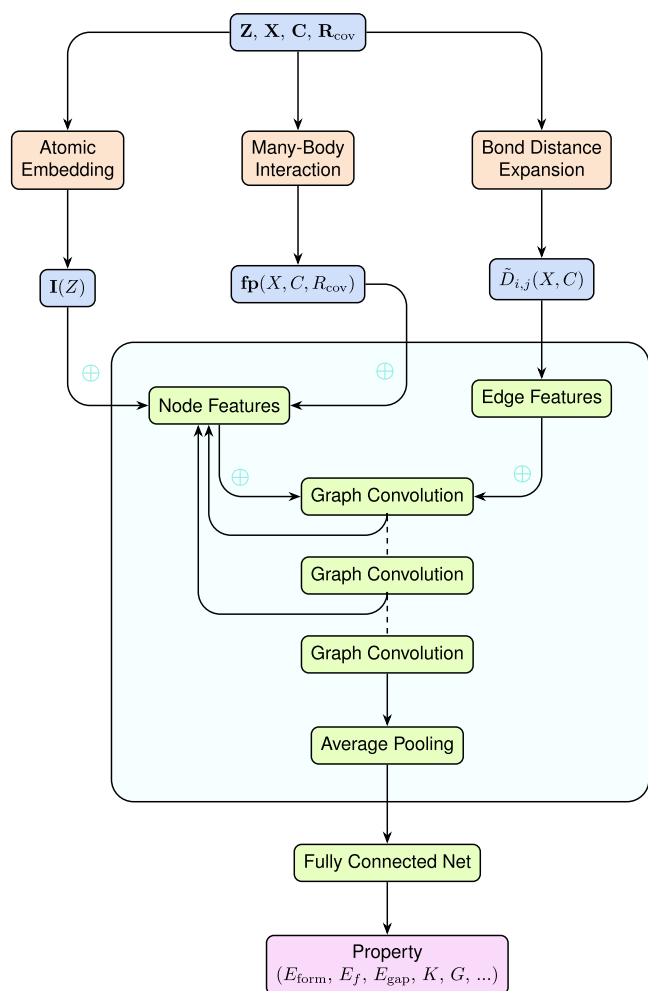


Figure 3. Workflow of EOSnet, which uses fractional coordinates ($\mathbf{X} \in \mathbb{F}^{N \times 3}$, where $\mathbb{F} = \{f \in \mathbb{R} : 0 \leq f < 1\}$), lattice vectors ($\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3] \in \mathbb{R}^{3 \times 3}$), atomic numbers ($\mathbf{Z}$), and covalent radii ($\mathbf{R}_{\text{cov}}$) as inputs. The $\tilde{D}_{i,j}$ denotes bond distance expansion using Gaussian filters, defined in eq 4. The process incorporates atomic embedding, many-body interaction, and bond distance expansion to generate node and edge features, which are processed through graph convolution layers and pooling to predict material properties.

convolutional layer, the node feature for atom i is updated by aggregating information from its neighboring atoms j through a message-passing mechanism. This process is defined as

$$\mathbf{n}_i^{(l+1)} = \mathbf{n}_i^{(l)} + \sum_{j,k} \sigma(\mathbf{W}_g^{(l)} \mathbf{m}_{(i,j)_k}^{(l)} + \mathbf{b}_g^{(l)}) \odot g(\mathbf{W}_m^{(l)} \mathbf{m}_{(i,j)_k}^{(l)} + \mathbf{b}_m^{(l)}) \quad (1)$$

$$\mathbf{m}_{(i,j)_k}^{(l)} = \mathbf{n}_i^{(l)} \oplus \mathbf{n}_j^{(l)} \oplus \mathbf{e}_{(i,j)_k}^{(l)} \quad (2)$$

Here, $\mathbf{m}_{(i,j)_k}^{(l)}$ is the message from atom j to atom i via the k -th edge, $\mathbf{n}_i^{(l)}$ denotes the node feature of atom i at layer l , $\mathbf{e}_{(i,j)_k}^{(l)}$ represent the k -th edge feature between atom i and j . And $\mathbf{W}_g^{(l)}$, $\mathbf{b}_g^{(l)}$ are the gate weight and bias at layer l , $\mathbf{W}_m^{(l)}$, $\mathbf{b}_m^{(l)}$ are the message weight and bias at layer l . Operator $\odot$ is the element-wise matrix multiplication, $\oplus$ is the vector concatenation, σ and g are independent nonlinear activation functions. EOSnet adopts a gated architecture similar to that introduced in the

CGCNN model.³¹ A simple convolution with a shared weight matrix $\mathbf{W}$ treats all neighbors identically, neglecting differences in their interaction strengths. By introducing a gate function σ and another nonlinear transformation g , it creates a multiplicative interaction that allows the network to selectively control how much information from each neighbor is incorporated.

The convolutional layers, as depicted in Figure 1, process the GOM-embedded node features, allowing the network to retain information about the many-body interactions throughout the depth of the model. After each message-passing step, node features are updated using gated convolutional layers. These layers ensure that relevant information from the neighbors is retained while irrelevant information is filtered out. This process is repeated for several layers, allowing the model to capture both local and global atomic interactions.

At the end of the graph convolutional layers, the node features are aggregated using a global pooling operation to predict the target property. In EOSnet, we use an average pooling function, which computes the mean of all node features in the graph, followed by a fully connected layer that maps the aggregated feature vector to the target property:

$$\text{Output} = \text{FCN} \left(\frac{1}{N_{\text{at}}} \sum_{i=1}^{N_{\text{at}}} \mathbf{n}_i^{(f*)} \right) \quad (3)$$

where N_{at} is the number of atoms in the crystal structure, and $\mathbf{n}_i^{(f*)}$ is the final node feature after the last convolutional layer. The fully connected layer (FCN) maps the aggregated node features to the target property, providing the final prediction. This averaging operation occurs between the hidden layers and the output layer in GCNN, as shown in Figure 1. After the graph convolution layers have generated hidden features for each atom, an average pooling step aggregates these atom-level embeddings into a single vector that represents the entire crystal. This pooled crystal-level feature is then passed to the fully connected layers to produce the final property prediction. The a_i in the input layer of Figure 1 represents the fingerprint-embedded atomic descriptor for the i -th atom in the crystal. Each a_i is derived from the GOM eigenvalues.

In addition to the GOM fingerprints used for node features, EOSnet also utilizes bond distance expansions to define the edge features between atoms. These edge features are expanded using Gaussian filters to ensure smooth and continuous representation of interatomic distances. The Gaussian distance expansion of the k -th bond is expressed as

$$\tilde{D}(i, j, k) = \exp \left(-\frac{(\|\mathbf{r}_{i,j}\| - d_k)^2}{\sigma_k^2} \right) \quad (4)$$

where $\|\mathbf{r}_{i,j}\|$ is the distance between atoms i and j , d_k is the center of the Gaussian filter and σ_k the scale parameter controlling the Gaussian width. These expanded bond distances provide additional geometric information for the convolutional layers, enabling the model to effectively learn from both node and edge features. The inclusion of these features is crucial for capturing short-range interactions, especially in systems where the geometric arrangement of atoms significantly impacts material properties. The workflow for this process is also detailed in Figure 3.

EOSnet is trained on a variety of material data sets, including both large (>20,000) and small (<2,000) data sets for material

property prediction. The model is trained using a supervised learning approach with an 80%–10%–10% train-validation-test split. Mean absolute error (MAE) is used as the loss function for regression tasks, such as predicting formation energy, total energy, and bandgap. For classification tasks, such as distinguishing between metals and nonmetals, cross-entropy loss is used. A dynamic class weight scheme is implemented to eliminate the effect of class imbalance in binary classification problems. The model is optimized using the Adam optimizer with a learning rate scheduler to dynamically adjust the learning rate during training. Gradient clip technique is used to prevent overfitting during training process.

The performance of the proposed EOSnet model was evaluated across various material property prediction tasks, including the prediction of total energy, formation energy, Fermi energy, band gap, bulk modulus, shear modulus, and a metal/nonmetal classification task. To benchmark the effectiveness of EOSnet, comparisons were made with several models, including CGCNN,⁵¹ MEGNet,⁴³ and M3GNet.³⁹ The summarized results are presented in Table 1. It is

Table 1. Comparison of the MAEs in Total Energy (E_{tot}), Formation Energy (E_{form}), Fermi Energy (E_f), Band Gap (E_g), Bulk Modulus (K_{VRH}), Shear Modulus (G_{VRH}) and Metal/Nonmetal Classification between EOSnet and Previous Works^a

	EOSnet ^b	CGCNN ⁵¹	M3GNet ³⁹	MEGNet ⁴³
E_{tot} (eV · atom ⁻¹)	0.034 (19,364)	0.072 (28,046)	0.035 (132,752)	-
E_{form} (eV · atom ⁻¹)	0.022 (131,240)	0.039 (28,046)	0.0195 (132,752)	0.028 (60,000)
E_f (eV)	0.295 (27,293)	0.363 (28,046)	-	-
E_g (eV)	0.163 (19,393)	0.388 (16,485)	0.183 (106,113)	0.33 (36,720)
K_{VRH} (log ₁₀ (GPa))	0.034 (5,000)	0.054 (2,041)	0.058 (10,987)	0.050 (4,664)
G_{VRH} (log ₁₀ (GPa))	0.072 (5,000)	0.087 (2,041)	0.086 (10,987)	0.079 (4,664)
Metal/Nonmetal Classifier	97.7% (19,393)	95.0% (16,458)	95.8% (106,113)	90.6% (55,391)

^aThe number of structures in the training data is in parentheses. ^bFor our model, the numbers in parentheses indicate the total numbers of structures before train-validation-test split. 19,364 E_{tot} data are from ASE⁵⁵ cubic perovskite data set.^{56–58} 27,293 E_f data and 19,393 band gap data are from Materials Project 2024 data set. 5,000 Voigt-Reuss-Hill (VRH) average bulk moduli, shear moduli are from Materials Project 2018 data set, and 131,240 E_{form} data are from Materials Project 2019 data set.

important to note that although all models listed in Table 1 were trained on data sets from the Materials Project,⁵¹ the sizes of the data sets and the evaluation metrics may differ. The primary focus of this comparison is not to claim that EOSnet is universally superior but to highlight its performance across various tasks. Specifically, we emphasize the advancements made by EOSnet in capturing many-body interactions to enhance the prediction accuracy of material properties. Studies show that the MAE of ML models generally decreases with the increase in the size of the training data set.^{39,43,52} Therefore, the performance of EOSnet is expected to improve further with larger training data sets in the future studies.

Figure 4a and 4b provide a detailed analysis of the EOSnet model performance in predicting formation energy and electronic band gap. The color gradient in these figures represents the density of data points, with brighter colors indicating regions of higher density. The tight clustering of predicted values around the diagonal line indicates that EOSnet effectively models the distribution of formation energy and band gap values, demonstrating robustness and reliability. An example demonstrating the strengths of EOSnet is its band gap prediction, achieving a low MAE of 0.163 eV. This performance is impressive comparing to other models, despite differences in the size of training data sets used. Band gap prediction is particularly challenging due to its sensitivity to many-body interactions, as well as electronic correlations within the material. Models that primarily rely on bond distances and basic geometric features often struggle to capture the nuanced effects that influence the electronic structure. The use of GOM-based fingerprints in EOSnet enables a more comprehensive representation of many-body interactions. This method captures not only the local interactions between an atom and its immediate neighbors but also the interactions among neighboring atoms themselves. For example, in semiconducting materials, the band gap, defined by the energy separation between valence and conduction bands, emerges from complex interplays between atomic arrangements, chemical bonding, and the electronic environment. While conventional GNNs often struggle to capture subtle electronic interactions and intricate orbital overlaps that determine band gaps, EOSnet effectively encodes these crucial quantum mechanical features.

The enhanced performance extends beyond band gap prediction to other properties sensitive to many-body interactions, such as bulk and shear moduli (Table 1). This broad success demonstrates the effectiveness of EOSnet approach to capturing complex atomic interactions in materials. The model also excels in classification tasks, achieving 97.7% accuracy in metal/nonmetal classification and 95.0% accuracy in predicting the dynamic stability of guest-atom substituted type-VII boron-carbide clathrates (Figure 4c and 4d). Particularly noteworthy is the model strong performance in stability classification despite limited training data, highlighting how the nonhuman-engineered atomic features can extract meaningful patterns even from small data sets.

In summary, we have introduced EOSnet (Embedded Overlap Structures for Graph Neural Networks), a new approach that enhances the predictive capabilities of graph neural networks in materials science by efficiently incorporating many-body interactions through GOM-based fingerprints. EOSnet provides a rotationally invariant and computationally efficient method to represent the full spectrum of atomic interactions without the need for explicit higher-order terms. Our extensive evaluations across various material property prediction tasks demonstrate the performance of EOSnet, particularly in predicting properties that are sensitive to many-body interactions such as electronic band gap and elasticity. Notably, EOSnet achieved a MAE of 0.163 eV in band gap prediction, surpassing the original CGCNN model and M3GNet. Same performance boost has been observed in the case of bulk modulus and shear modulus. This indicate embedding GOM fingerprints into node features can indeed helps the GNN model understand these properties better with the essential information on atomic environments, including

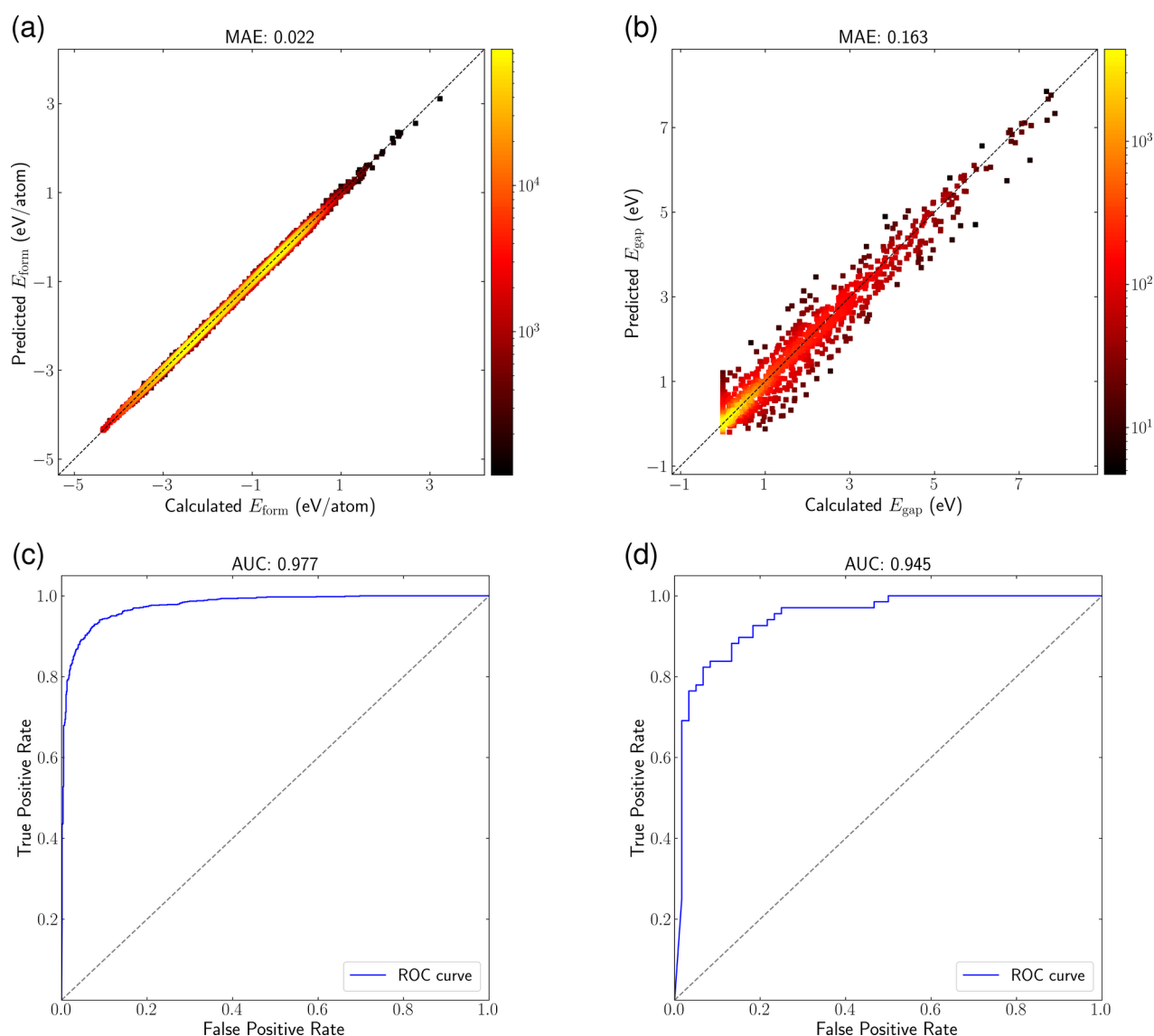


Figure 4. Evaluation for regression and binary classification tasks on the performance of our EOSnet model. (a) Parity plot for formation energy predictions on 131,240 data points from the Materials Project, with MAE of 0.022 eV/atom. (b) Parity plot for the prediction of electronic band gap (E_g) using 19,393 data points from the Materials Project, showing an MAE of 0.163 eV. (c) ROC curve for metal/nonmetal classification, achieving an AUC of 0.977. (d) ROC curve for dynamically stable/unstable classification on 1,335 guest-atom-substituted type-VII boron-carbide clathrates ($\text{MB}_{6-x}\text{C}_x$, x from 1 to 5), achieving an AUC of 0.945. The clathrate stability is determined using phonon frequency from VASP^{53,54} calculations.

the collective behavior of neighboring atoms and the strength of their orbital overlaps. This improved performance is especially important for applications in material discovery, making it a valuable tool for the design of materials with optimized electronic, mechanical, and thermal properties. While EOSnet shows performance improvements, further work could include expanding the model to handle more diverse data sets and incorporating additional geometric or attention mechanisms to capture long-range interactions more effectively. Additionally, exploring its applicability to other domains, such as catalysis or battery materials, could open new avenues for material discovery.

■ ASSOCIATED CONTENT

Data Availability Statement

The source code and data for the EOSnet model are available at the following GitHub repository: <https://github.com/Rutgers-ZRG/EosNet>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.4c03179>.

Additional details on the model architecture, and parameters for the model configuration. It includes additional figures that further illustrate the model performance across various material property prediction tasks. eq 1 shows Gaussian-type orbital overlap elements for Gaussian Overlap Matrix (GOM); Figures S1–S3 showing the comparison of performance of the EOSnet model with and without GOM Fingerprints for

predicting various properties; Figure S4 showing the evaluation of the EOSnet model performance on formation energy predictions across varying data set sizes; Figure S5 is the histogram showing the distribution of elements per unit cell in different data set sizes; Table S1 summarizing EOSnet model parameters and training settings (PDF)

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Notes

The authors declare no competing financial interest.

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